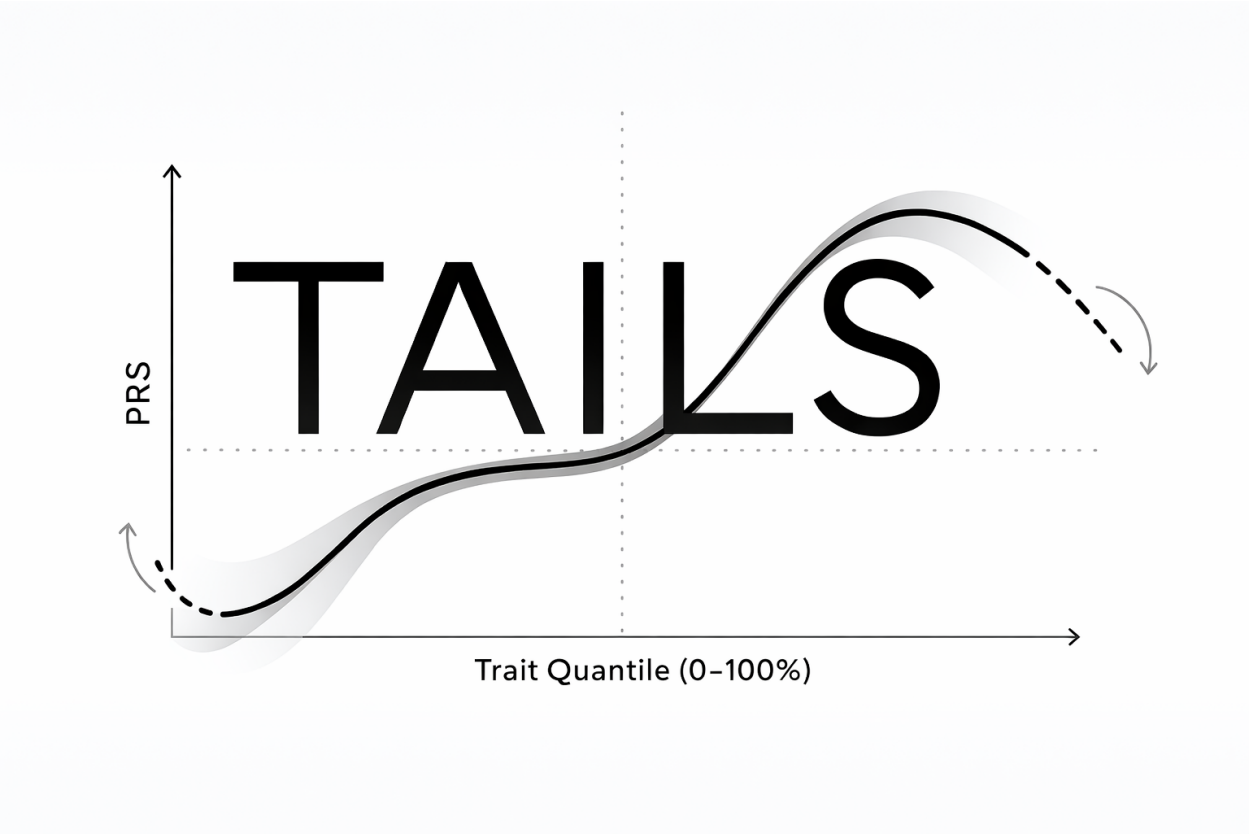
TAILS pipeline manual



1 Overview

The TAILS package has two command-line stages:

1. `tails qc`: merge per-trait QC summaries, inspect available variables, apply filters, and remove highly similar traits.
2. `tails gen`: generate manuscript figures, extended-data figures, and CSV summary tables from the QC-passed trait set together with packaged analysis outputs.

The intended workflow is sequential: the QC stage writes a filtered trait file, and the figure-generation stage reads that QC-produced file. In the packaged examples, the standard handoff file is:

```
output_qc/qc.pass.txt
```

2 Package layout

`code/tails.py`	Main command-line entry point.
`code/src/`	Pipeline source code
`data/`	Bundled trait, QC, and figure-generation inputs.
`docs/`	Documentation.
`run_all.sh`	Example end-to-end script.

3 Requirements

3.1 System requirements

- Linux or macOS shell environment
- Python 3.10+ recommended
- `bash`

3.2 Python requirements

The figure-generation code checks for the following Python modules:

- `matplotlib`
- `numpy`
- `scipy`
- `pandas`
- `statsmodels`
- `pyreadr`

`pyreadr` is only needed for the simulation-figure component of `tails gen all`; however, the simplest setup is to install the full environment up front.

Example installation:

```
python3 -m pip install matplotlib numpy scipy pandas statsmodels pyreadr
```

4 Top-level commands

From the package root:

```
python3 code/tails.py qc
python3 code/tails.py gen
```

Useful help commands:

```
python3 code/tails.py -h
python3 code/tails.py qc -h
python3 code/tails.py qc load -h
```

```
python3 code/tails.py qc filter -h
python3 code/tails.py qc run -h
python3 code/tails.py gen help
```

Note that `tails gen -h` is minimal. The generation targets are exposed through `tails gen help` and through the parsing logic documented below.

5 QC stage

The QC stage can be run in two ways:

1. in two steps with `qc load` followed by `qc filter`; or
2. in one step with `qc run`.

5.1 QC load

```
python3 code/tails.py qc load \
  --in data/namedTraits.txt \
  --qcFiles data/qc/txt/*.txt \
  --out output_qc/qc \
  --showVariables
```

This command reads the input trait list, merges the QC files, and writes:

- `output_qc/qc.merged.txt`
- `output_qc/qc.missing.txt` (if required QC values are absent and `-allowMissing` was not used)

QC load arguments

`-in`	Input trait file. Required.
`-qcFiles`	One or more QC summary files. Required.
`-out`	Output prefix. Required. This is a prefix, not a directory.
`-showVariables`	Print the available QC/filter variable names and example filters.
`-allowMissing`	Keep traits even when one or more QC files do not supply values.
`-silent`	Suppress progress output.
`-verbose`	Verbose flag exposed by the CLI.

5.2 QC filter

```
python3 code/tails.py qc filter \
  --in output_qc/qc.merged.txt \
  --out output_qc/qc \
  --applyFilters "sampleSize-target>50000 h2>0.05 snpmax<0.02 dist-skew<|2| uniq50>2 popQC-pass=
  --dupeFile data/qc/gcorr.mat \
  --configFile data/qc/config.txt
```

This command applies user-specified filters or the built-in standard filter set, then optionally applies a custom keep/remove configuration and duplicate removal.

Outputs are typically:

- `output_qc/qc.pass.txt`
- `output_qc/qc.fail.txt`

QC filter arguments

`-in`	Input file, typically the merged file from `qc load`. Required.
`-out`	Output prefix. Required.
`-applyFilters`	Space-separated filter string. Mutually exclusive with `-useStandardFilters`.
`-useStandardFilters`	Apply the built-in standard filter string. Mutually exclusive with `-applyFilters`.
`-configFile`	Optional custom keep/remove configuration file.
`-dupeFile`	Optional duplicate-correlation file used for deduplication.
`-dupeCutoff`	Deduplication threshold. Default: `0.99`.
`-dupeTieBreak`	Tie-breaking rule passed to the deduplication logic. Default: `N`.
`-showVariables`	Show available variables before or during filtering.
`-silent`	Suppress progress output.
`-verbose`	Verbose flag exposed by the CLI.

Standard QC filters

The built-in standard filter set used by `-useStandardFilters` is:

```
sampleSize-target>50000 h2>0.05 snpmax<0.02 dist-skew<|2| uniq50>2 popQC-pass=True
```

Filter syntax

Filters are supplied as a space-separated string. Each token has the form:

```
VARIABLE OPERATOR VALUE
```

with no spaces inside a token. Supported operators are:

```
<   >   <=   >=   =   !=
```

Examples:

```
h2>0.05
sampleSize-target>50000
popQC-pass=True
uniq50>2
```

The QC code also supports symmetric absolute-value style thresholds using the `|value|` syntax. For example:

```
dist-skew<|2|
```

means `-2 < dist-skew < 2`.

When in doubt, run `qc load -showVariables` to see the variable names that this particular QC input bundle exposes.

Config file behavior

The optional configuration file passed through `-configFile` can specify explicit keep/remove actions. The parser recognizes `KEEP` and `REMOVE` rules and allows pre/post timing rules. The packaged example uses:

```
data/qc/config.txt
```

5.3 QC run

`qc run` combines `qc load` and `qc filter` in one command.

Example:

```
python3 code/tails.py qc run \
  --in data/namedTraits.txt \
  --qcFiles data/qc/txt/*.txt \
  --out output_qc/qc \
  --useStandardFilters \
  --dupeFile data/qc/gcorr.mat \
  --configFile data/qc/config.txt
```

Its arguments are the union of `qc load` and `qc filter`.

6 Figure-generation stage

The figure-generation stage reads the QC-passed trait file together with bundled analysis files:

- `-vals`: one or more `.vals` files
- `-pts`: one or more `.pts` files
- `-simPath`: optional simulation directory used by the simulation figures

Basic form:

```
python3 code/tails.py gen WHICH --in QC_PASS --vals ... --pts ...
```

6.1 Figure-generation arguments

`WHICH`	Generation target such as `all`, `fig2`, `xtd4`, or `csv`. Required positional argument.
`-in`	QC-passed trait file. Required.
`-vals`	One or more `.vals` files. Required.
`-pts`	One or more `.pts` files. Required.
`-simPath`	Simulation-results directory. Needed for the simulation figure set.
`-dpi`	Figure DPI. Default: `500`.
`-indexTraits`	Four integer trait IDs used as index traits. Default: `30810 30070 30020 20015`.
`-out`	Output directory or output prefix for figures/tables. Default: empty string.
`-silent`	Suppress progress output.

6.2 Supported generation targets

The generation parser recognizes the following groups.

Aggregate targets

`all`	Generate all main figures, all extended-data figures, the CSV outputs, and the simulation figure set.
`main` / `mains` / `main-figs` / `figs`	Generate all main figures (1–5).
`xtd` / `ext` / `extended-data` / `sups`	Generate all extended-data figures (1–10).
`csv` / `csvs` / `tabs` / `tables` / `excel`	Generate the CSV table outputs only.
`sim` / `sims`	Generate the simulation figure set only.

Single-target forms

`main1` to `main5`	Individual main figures.
`fig1` to `fig5`	Accepted aliases for the main-figure naming convention used in examples.
`xtd1` to `xtd10`	Individual extended-data figures.
`sup1` to `sup10`	Accepted supplementary aliases for extended-data figures.

Range forms

The parser also accepts ranges such as:

```
main1-3
fig1-3
xtd2-5
sup2-5
```

Main-figure ranges must stay within 1–5, and extended-data ranges must stay within 1–10.

6.3 Main figure-generation example

```
python3 code/tails.py gen all \
  --in output_qc/qc.pass.txt \
  --vals data/trait_data/vals/*.vals \
  --pts data/trait_data/pts/*.pts \
  --simPath data/trait_sims/ \
  --out output_figs/
```

6.4 Compiling the extended-data PDF

After figure generation, the extended-data LaTeX source can be compiled with:

```
cd xtended_latex_src
./compileExtendedDocument.sh ../output_figs/
cd ..
```

This writes:

```
xtended_latex_src/extendedData.pdf
```

7 End-to-end packaged workflow

A standard packaged workflow is:

```
mkdir -p output_qc output_figs

python3 code/tails.py qc run \
  --in data/namedTraits.txt \
  --qcFiles data/qc/txt/*.txt \
  --out output_qc/qc \
  --useStandardFilters \
  --dupeFile data/qc/gcorr.mat \
  --configFile data/qc/config.txt

python3 code/tails.py gen all \
  --in output_qc/qc.pass.txt \
  --vals data/trait_data/vals/*.vals \
```

```
  --pts data/trait_data/pts/*.pts \
  --simPath data/trait_sims/ \
  --out output_figs/

cd xtended_latex_src
./compileExtendedDocument.sh ../output_figs/
cd ..
```

8 Outputs

Typical outputs include:

- `*.merged.txt`, `*.missing.txt`, `*.pass.txt`, and `*.fail.txt` from the QC stage
- main-figure PDFs
- extended-data figure PDFs
- CSV table exports
- `xtended_latex_src/extendedData.pdf`

9 Documentation map

`README.md`	Top-level package overview and quick-start instructions.
`docs/README.qc`	Quick QC commands.
`docs/README.gen`	Quick figure-generation commands.
`docs/tailsManual.pdf`	Full manual.